PRODUCTION-DEPLOYED MULTI-TENANT BIOMETRIC IDENTITY SAAS · ONE HOSTED LOGIN BUTTON

<FIVUCSAS />

Face & Identity Verification using *Cloud-based SaaS* Models

TEAM · DEPARTMENT OF COMPUTER ENGINEERING

Ahmet Abdullah Gültekin ahmetabdullahgultekin@gmail.com github.com/ahmetabdullahgultekin

Ayşe Gülsüm Eren ayse.gulsum@marun.edu.tr aysegulsumeren@gmail.com github.com/aysegulsum

Ayşenur Arıcı aysenurarici@marun.edu.tr aysenurarici@hotmail.com github.com/Aysenur15

SUPERVISOR Assoc. Prof. Dr. Mustafa Ağaoğlu

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01 Introduction

PROBLEM · WHAT WE BUILT · MAIN GOAL · OBJECTIVES

02 The Drop-in Identity Button

E-DEVLET & RECAPTCHA MENTAL MODEL · SIDE-BY-SIDE · 3 USE-CASES

Three buttons, one mental model

- T.C. e-Devlet ile Giriş Yap — TURKIYE.GOV.TR · PUBLIC SECTOR SSO
- I'm not a robot — reCAPTCHA — GOOGLE.COM · BOT PROTECTION DROP-IN
- </> Sign in with FIVUCSAS — VERIFY.FIVUCSAS.COM · BIOMETRIC SAAS DROP-IN

Side-by-side comparison

ASPECT	E-DEVLET	RECAPTCHA	FIVUCSAS
NFC document	—	—	T.C. + ICAO 9303 (mobile) + Web NFC (simple cards)
Integration	Gov approval	Public site-key	Public OIDC + drop-in button
Self-host	—	—	Docker Compose · GPU-less
Open source	—	—	MIT (spoof-detector) · OSS roadmap

03 System Architecture

HEXAGONAL / DDD · 7 CONTAINERS · HETZNER CX43

Client Web · KMP · Widget	→	**Traefik v3.6** TLS · JWT · rate-limit	→	**Identity Core + Bio-Processor** Spring · FastAPI · hexagonal
PostgreSQL 17 pgvector · RLS · 60 migrations		**Redis 7.4** cache · OTP · puzzle nonces		**Hetzner CX43** 8 vCPU · 16 GB · no GPU

04 Face Recognition Pipeline — 8 Stages

DETECTION → QUALITY → LANDMARK → ALIGN → EMBED → ENCRYPT → INDEX → MATCH

Server-authoritative pipeline in `biometric-processor/` (FastAPI, Python 3.12). Every stage is replaceable — backbone, detector and similarity all swap via DI in `app/infrastructure/ml/factories/`. Verification (1:1) compares against the enrolled vector with cosine distance; identification (1:N) returns pgvector ANN top-*k*. Aged embeddings (> 2 years) use a more permissive threshold.

- 01 · DETECT + QUALITY — **DeepFace 0.0.98** — opencv · anti-spoof θ=0.5 · Laplacian blur ≥100 · min 80px · occlusion CIE-Lab — **quality ≥ 70**
- 02 · LANDMARK + ALIGN — **MediaPipe Tasks** — 468-point FaceLandmarker · DeepFace eye-line affine — **468 + 10 iris**
- 03 · EMBED + ENCRYPT — **Facenet-512 · Fernet** — 512-D L2-normalised · AES-128-CBC + HMAC-SHA256 at rest — **dim = 512**
- 04 · INDEX + MATCH — **pgvector · cosine** — HNSW m=16/ef=64 · 1:N ANN top-k · aged-embedding θ=0.38 — **θ = 0.45**

05 NFC Document Subsystem — Two Rails

NATIVE MOBILE (ICAO 9303) · WEB NFC (SIMPLE CARDS) · CROSS-CHECK TO LIVE SELFIE

Two rails, one verifier. Rail A — native mobile NFC: T.C. ID + ICAO 9303 passports + any BAC-chip document over Android IsoDep + Bouncy Castle. **Rail B — Web NFC:** ISO 14443 simple cards (loyalty, transit, NTAG2xx) directly from Chrome on Android.

T.C. Yeni Kimlik Kartı

The contactless ISO/IEC 14443 chip lives on the back of the T.C. ID card; on passports it is in the cover. Common ICAO AID `A0 00 00 02 47 10 01`. The same BAC key derivation (SHA-1 from M
reused for TD3 passports and every other ICAO 9303 document carrying the same chip.

Standard: ICAO Doc 9303 (Machine Readable Travel Documents)
Spec ref: `practice-and-test/PASSPORT_NFC_ROADMAP.md:8`

A. Protocols implemented

- **BAC (Basic Access Control)** — SHA-1 key derivation from MRZ, identical for Turkish T.C. ID and TD3 passports. `PASSPORT_NFC_ROADMAP.md:670-690`
- **PACE** (Password Authenticated Connection Establishment) — optional for newer passports, scoped not implemented. `:164-170`
- **Active Authentication · Chip Authentication** — *[future work]*
- **EF.SOD** validator — 60% complete; CSCA full chain pending. `:383-395`
- **ICAO check-digit** validation: `biometric-processor/tests/unit/test_nfc_mrz_route.py:47-88`

B. Data Groups parsed

- **DG1 — MRZ**: `.../TurkishEidNfcReader/.../util/Dg1Parser.kt`
- **DG2 — Facial image** (JPEG2000 / JP2): `Dg2Parser.kt:1-50`
- **EF.COM** — data-group presence list
- DG3 / DG4 / DG11 / DG13 / DG14 / DG15: *not implemented (roadmap §5.1)*

06 spoof-detector — Hybrid PAD, Session Verdict, In-Browser

SUBMODULE · MIT · PAPER DRAFT · IBETA L1 SUBMISSION · AMISPOOF.FIVUCSAS.COM

Measured results (paper, bootstrap 95 % CI on 100 stratified resamples)

DATASET / REGIME	PIPELINE	VALUE	95 % CI	SOURCE
CASIA-FASD AUC (zero-shot, N = 2 408)	minifasnet_only	**0.9452**	[0.9366, 0.9560]	07_results.md:15
Per-frame latency (Hetzner CX43 CPU)	hybrid	**63.0 ms**	p99 = 117.8 ms	:110
Browser (desktop Chrome, WebGPU)	browser bundle	**25 – 30 fps**	—	BROWSER_READINESS.md:4
ISO/IEC 30107-3 (in-house scripted)	session	**Grade C**	BPCER 0 % · APCER 30 % · ACER 15 %	README.md:74-84

07 The Biometric Puzzle — 23 Micro-Challenges

14 FACE + 9 HAND-GESTURE · SERVER-NONCED · REPLAY-PROOF · CANONICAL ENUM BIOMETRICPUZZLEID

23 canonical micro-challenges (14 face + 9 hand). Server emits a random UUID-nonced sequence of 2-7 actions per attempt; client streams only landmark vectors. Pre-recorded, AR-filter, deepfake, replay all fail — sequence unpredictable, server-immutable.

14 face + 9 hand challenges
blink · smile · open mouth · turn L/R · look up/down · raise brows · close eye · nod · shake · finger count · wave · flip · tap · pinch · peek-a-boo · math · shape-trace · template-trace · hold-pose

08 Ten Composable Authentication Factors

NIST 800-63B · KNOW / HAVE / ARE / SHOW · TENANT JSON, NOT JAVA

∑ 10 Per-tenant MFA composition · no backend code required

Password	KNOW	BCrypt-12
Email OTP	HAVE	6-digit · 5 min
SMS OTP	HAVE	Twilio · 6-digit
TOTP	HAVE	RFC 6238 · 30 s
QR Code	HAVE	cross-device
Face	ARE	Facenet-512 + Puzzle
Voice	ARE	Resemblyzer · 256-D
Fingerprint	ARE	WebAuthn platform
Hardware Key	HAVE	FIDO2 · WebAuthn
NFC Document	SHOW	ICAO 9303 · BAC

10 Subdomain Ecosystem

9 SUBDOMAINS · 1 QR · ALL BEHIND TRAEFIK V3.6

One QR · all links. A single QR code on this poster points to a **link navigator page we own**: `links.fivucsas.com`. We do not depend on a third-party shortener — our own static subdomain hosts every project subdomain, the team (GitHub, LinkedIn, portfolio, email), poster artefacts (HTML, PDF, PNG, text brief) and shortcuts to live surfaces. Hostinger static (no Docker), the same pattern as bys-demo / amispoof / landing.

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Scan to open the FIVUCSAS hub

A single QR for landing + hosted-login + admin + docs + demo + spoof-detector + status — every surface under one short link. Scan from any phone and jump to the surface you need.

https://fivucsas.com/links

REPOSITORIES

github.com/Rollingcat-Software/fivucsas (private)
github.com/Rollingcat-Software/spoof-detector (MIT)
6 git submodules · 1 self-hosted runner

LIVE SURFACE

fivucsas.com · verify.fivucsas.com · demo.fivucsas.com
api.fivucsas.com · app.fivucsas.com · docs.fivucsas.com
amispoof.fivucsas.com · status.fivucsas.com

CONTACT

ahmetabdullahgultekin@gmail.com
ayse.gulsum@marun.edu.tr · aysegulsumeren@gmail.com
aysenurarici@marun.edu.tr · aysenurarici@hotmail.com
Supervisor · ağaoğlu @ marmara.edu.tr